

Module E (WIP): a neutral null for ancestry sorting, and what the empirical population structure already rules out

Work in progress — July 25, 2026

Abstract

This is a working note on Module E: a recombination-matched, haplodiploid *neutral null* for the 20 replicate *F. polycтена* $\times$ *F. aquilonia* hybrid populations, built to test whether the directional ancestry sorting seen in Modules A–C exceeds drift. The null conditions on real phased parental haplotypes, uses the empirical recombination map, is haplodiploid, and contains no selection. The central design choice is to stratify markers by DiagnosticIndex (DI): low-DI markers cannot be driven by ancestry sorting and so serve as an *internal demographic control* in the same analysis that tests the high-DI markers. Two results now stand largely on the empirical data alone, independent of the simulation calibration. First, among-population differentiation scales steeply with DI in the data (F_{ST} : ~ 0 at neutral markers $\rightarrow 0.31$ at diagnostic markers) whereas neutral drift, being DI-blind, is flat. Second, decomposing the diagnostic differentiation shows that a single per-population admixture proportion (“variable founding admixture”) explains only $\sim 6\%$: the differentiation is 78% *locus-specific*. Because neutral drift is excluded (neutral $F_{ST} \approx 0$) and variable admixture is excluded (6%), what remains is a large, locus-specific, *divergent* ancestry signal standing against a near-zero neutral expectation. A wrinkle worth flagging: this signal is divergent (populations differing at diagnostic loci), which is distinct from — though compatible with — the parallel directional sorting of Module A. The definitive test, a p -value from the sorting statistic run through the calibrated null, is in progress.

1 The null model

Rather than simulating the parental species’ history, we *condition* on it: founders are drawn from real phased parental haplotypes (15 diploid females per species, imputed and phased within species with Beagle 5.4; pool of 30 *aquilonia* haplotypes and 13 *polycтена* diploid females), the mutation rate is zero, and only the post-contact hybrid phase is simulated. This inherits true allele frequencies, true between-species divergence, and true within-species haplotype structure. The engine is SLiM 5, non-Wright–Fisher, haplodiploid (haploid males from unfertilised eggs, spermathecal storage, Pamilo mating, age-structured life table), over the 26 assembled chromosomes with their empirical cM maps (converted with `scale`= 10^{-8}). All selection is removed, so drift and recombination on the empirical map are the only forces. Each parameter set runs as 20 independent demes sampled every 20 generations to generation 160.

Requirements for a valid comparison. Three points are load-bearing and worth recording. (i) **Sample-size matching:** composite r^2 and the sorting classification are both sample-size sensitive, so simulated demes are subsampled to the empirical per-population sizes (3–20 diploids). (ii) **Symmetric treatment:** any filter applied to the data is applied to the null (“sorted” *is* high differentiation, so filtering the data alone would move it toward the null mechanically). (iii) **F_{ST} estimator:** with per-population n of 3–20, the Nei estimator is upward-biased by $\mathcal{O}(1/2n)$; the Hudson estimator corrects this and is used for the structure results below. This correction matters (Section 3).

Neutral alternatives excluded as controls. Before interpreting any excess we quantified and excluded several neutral explanations. *Founder heterogeneity* (populations founded from different local sub-pools): even the absolute ceiling — each deme founded from a single *aquilonia* haplotype and a single *polycytena* female — gives an at-founding among-population F_{ST} of only 0.090, because at diagnostic markers both species are near-fixed for opposite alleles and a 50/50 admixture washes out within-species structure. *Colony (nest) sampling*: measured within-nest excess genotype correlation is only 0.047 (polygyny), so sibling structure is a minor effect. *Mixed population ages*: age mixtures interpolate along the LD-versus-drift trade-off and match no summary. These are recorded so they need not be re-litigated.

2 Differentiation and sorting scale with DI in the data, not under drift

Markers were binned by DI over the parent-MAF ≥ 0.15 universe (40,000 markers, 4,000 per bin across ten fixed DI bins). Figure 1 and Table 1 give the result: observed among-population F_{ST} rises from the neutral background to the most ancestry-informative markers, while the neutral null is essentially flat. The fraction of sorted loci shows the same pattern more sharply, from a ~ 1.5 -fold excess at low DI to >100 -fold at high DI. Because drift does not distinguish informative from uninformative markers, a DI-scaled discrepancy cannot be produced by demographic misspecification — the comparison is internally controlled because both datasets use the same markers, MAF gate, and frequency spectrum. (Table 1 reports Nei F_{ST} ; the absolute neutral value there, 0.05, is the small-sample bias corrected in Section 3, but the *scaling* is unaffected.)

Table 1: Observed vs simulated statistics by DI bin at the anchor-matched neutral cell. “sorted” is the fraction of loci classed as directionally sorted.

DI bin	F_{ST} obs	F_{ST} null	π obs	π null	sorted obs	sorted null
≤ -90	0.051	0.191	0.448	0.381	0.026	0.021
$(-75, -60]$	0.303	0.183	0.260	0.314	0.350	0.233
$(-40, -30]$	0.305	0.167	0.277	0.364	0.260	0.072
$(-25, -20]$	0.310	0.160	0.297	0.398	0.150	0.015
> -15	0.315	0.156	0.304	0.415	0.116	0.000

3 Population structure: neutral ≈ 0 , diagnostic strong

This is the pivotal result. At *strictly* neutral markers (DI ≤ -90 , 13,978 markers), the pairwise Hudson F_{ST} among the 20 populations is essentially **zero** (median 0.000, max 0.012), and the neutral PCA is a single overlapping cloud with no population structure (Fig. 2, left and middle). At diagnostic markers (DI > -25) the same populations are **strongly structured**, $F_{ST} = 0.31$, resolving into distinct clusters (Fig. 2, right).

Two consequences follow. First, the earlier “neutral $F_{ST} \approx 0.05$ ” was small-sample bias in the Nei estimator; the Hudson estimate is ≈ 0 . The same bias inflated the *simulated* Nei values equally, so the sim-versus-data gap in Table 1 survives, but the true neutral differentiation is ≈ 0 . Second, and more importantly, this dissociation cannot be drift: drift is DI-blind, so it cannot differentiate diagnostic markers while leaving neutral markers undifferentiated. The populations do not behave as 20 independently drifting demes at neutral loci (independent foundings accumulate $F_{ST} \approx t/2N_e > 0$); they look effectively panmictic / very-large- N_e / connected at neutral loci, while differing strongly in ancestry. A Mantel-style test confirms the two are dissociated: Spearman correlation between pairwise neutral F_{ST} and pairwise |ancestry difference| is only 0.08.

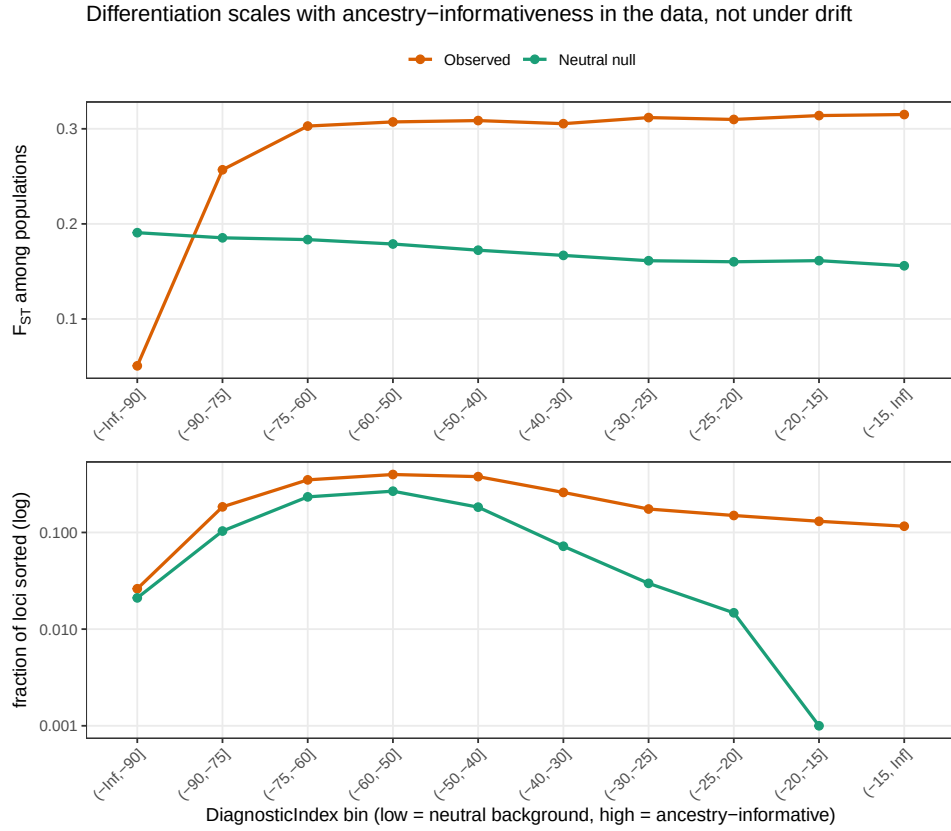


Figure 1: Among-population differentiation (top) and the fraction of sorted loci (bottom, log scale) against DI bin, observed vs neutral null. Neutral drift is DI-blind and produces an essentially flat profile; the data concentrate both differentiation and sorting in ancestry-informative markers.

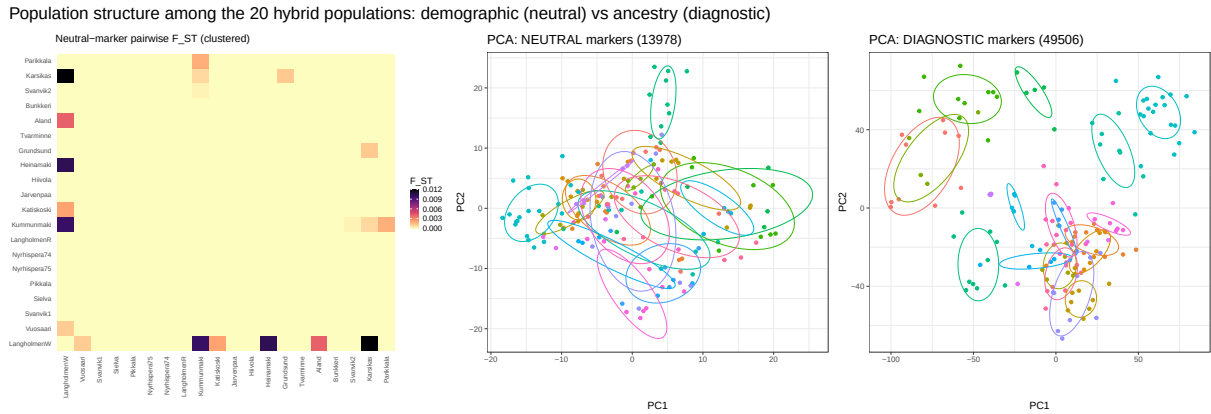


Figure 2: Structure among the 20 hybrid populations. Left: pairwise neutral-marker ($DI \leq -90$) F_{ST} , Hudson estimator, clustered — essentially zero everywhere. Middle: individual PCA on neutral markers — one overlapping cloud. Right: individual PCA on diagnostic markers — strong, structured differentiation.

4 The diagnostic differentiation is locus-specific, not variable admixture

If the populations were simply founded at different aquilonia:polycytena ratios, each would carry a single ancestry proportion a_j and its diagnostic allele frequencies would follow $p_{ij} = f_i^{\text{pol}} + a_j \Delta p_i$ (rank-1). We fit this per population and decomposed the among-population variance at diagnostic markers into: sampling (binomial), the single ancestry axis, and a genuine locus-specific residual. Of the observed diagnostic $F_{ST} = 0.313$:

component	F_{ST} (fraction)
sampling noise	0.049 (16%)
single ancestry axis (variable admixture)	0.019 (6%)
locus-specific residual	0.245 (78%)

Per-population best-fit ancestry does vary (0.35–0.65, SD 0.077; Fig. 3 top), but that single knob explains only $\sim 6\%$ of the differentiation. Variable founding admixture is therefore rejected as the explanation.

Combining the two structure results: drift is excluded (neutral $F_{ST} \approx 0$) and variable admixture is excluded (6%), so the 0.245 locus-specific diagnostic differentiation is neither. This is where the internal control pays off: a neutral null calibrated to reproduce neutral $F_{ST} \approx 0$ has essentially no drift and therefore predicts ≈ 0 diagnostic differentiation, making the observed 0.245 *almost entirely excess over neutral*.

An honest wrinkle: divergent, not parallel. The among-population *variance* decomposed here measures *divergence* — populations differing at diagnostic loci. Parallel sorting (all populations fixing the same allele at a locus, Module A’s DI-governs-direction result) produces agreement, i.e. *low* among-population variance, and lives in the *mean* shift rather than this residual. So the 0.245 is a *divergent* locus-specific signal, and the residual PCA is correspondingly diffuse and high-dimensional (Fig. 3 bottom; PC1/PC2 $\approx 9\%$ each, though small per-population n confounds this view). Divergent ancestry differentiation without drift points toward local/divergent processes (the extrinsic/climate arm, Module C) more than a single shared intrinsic incompatibility axis. It does not contradict Module A — a parallel mean direction and divergent variance can coexist — but it says the *differentiation among populations* is dominated by locus-specific divergence.

5 Supporting diagnostic: the LD landscape

Local LD (ld_w , median r^2 in a 100 kb window) along the genome (Fig. 4) shows that the null reproduces the recombination-driven LD *landscape* well — peaks co-localise on the same chromosomes, per-chromosome Spearman 0.94–0.95, robust across generations — but carries a diffuse genome-wide LD *floor* the data lack (mean ~ 0.17 – 0.21 vs 0.09), decaying only slowly from gen20 to gen160. A within-group analysis (removing the between-population component) showed the founders themselves carry within-species LD matching the empirical within-population LD, so the excess floor is not founder-inherited; it is the between-population (Wahlund) component, i.e. the same over-differentiation (too-small N_e) seen in the F_{ST} analysis. This independently corroborates that the null over-differentiates the demes and motivates the large- N_e calibration.

6 Where this stands, and next steps (*WIP*)

Two demographic explanations for the low-neutral / high-diagnostic dissociation are now excluded from the empirical data alone: *drift* (neutral $F_{ST} \approx 0$, and drift is DI-blind) and *variable founding admixture* (6% of the diagnostic differentiation). What remains is a large ($F_{ST} \approx 0.245$), locus-specific, *divergent* ancestry signal against a near-zero neutral expectation. The simulation’s role

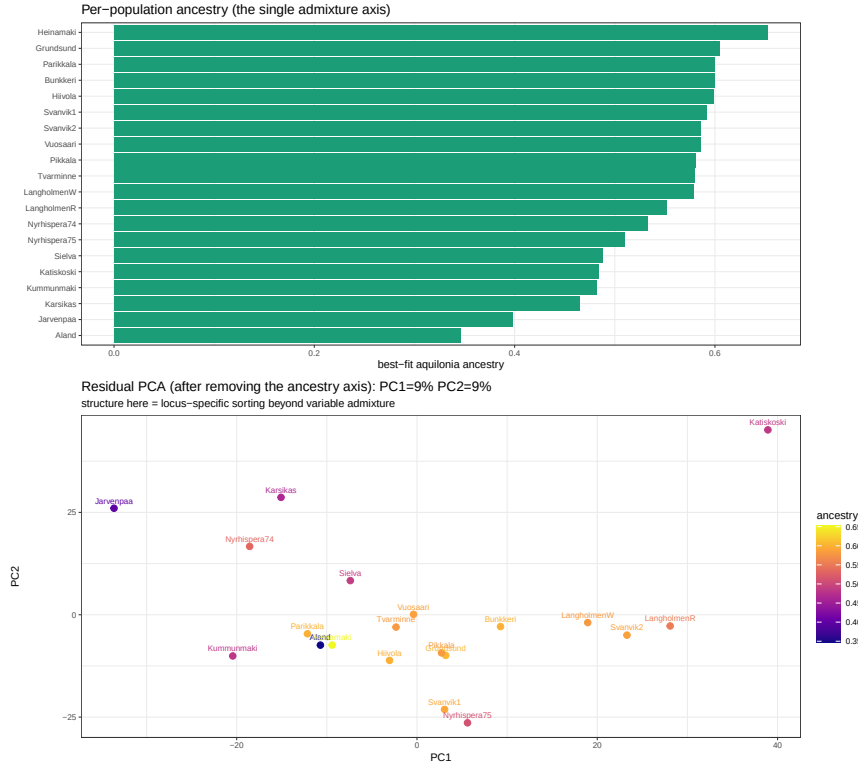


Figure 3: Top: per-population best-fit aquilonia ancestry (the single admixture axis), 0.35–0.65. Bottom: PCA of the residual after removing that axis; structure here would be locus-specific sorting beyond variable admixture. It is diffuse and high-dimensional (sampling-confounded at these sample sizes).

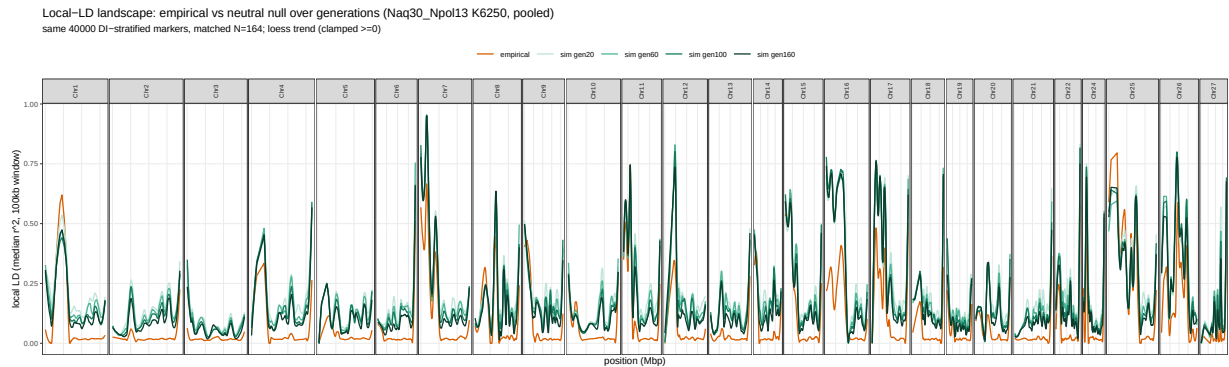


Figure 4: Local LD along the genome, empirical vs neutral null over generations ($K=6250$, pooled, matched N). The recombination-driven landscape is reproduced (peaks co-localise); the null carries a diffuse excess floor between peaks — the between-population (Wahlund) component of the over-differentiation.

is now sharply defined: calibrate the null to reproduce neutral $F_{ST} \approx 0$ (the very-large- N_e / effectively-panmictic-at-neutral regime), at which point the diagnostic excess is read against ≈ 0 .

1. **Calibrated-null p -value.** Run the sorting statistic (`parallelism_stats`) through the null calibrated on the neutral markers, extending carrying capacity upward until neutral $F_{ST} \approx 0$. This puts a p -value on the excess and separates the parallel (mean) from the divergent (variance) components properly. *Higher- K simulations are running on a larger machine.*
2. **Divergent vs parallel.** Characterise the residual structure with individual-level data (controlling sampling), and test whether it aligns with climate (reconciling with Modules A and C: parallel direction vs divergent differentiation).
3. **Non-independence.** Neutral $F_{ST} \approx 0$ is formally degenerate between enormous N_e and gene flow / shared ancestry among “populations”; extra within-event sampling and demographic modelling (cf. Nouhaud et al.) would separate these, and it bears on how many independent replicates the sorting test really has.

All analyses run from a 19.5 MB compact empirical bundle (replacing the 1 GB dataset); scripts and the portable simulation pipeline are in `moduleE.slim/` in the repository.